

Syllabus for

School year 2025/2026 >

DAT410 - Design of AI systems

Design of AI systems

The course syllabus was approved on 2021-02-26 by the program manager (or equivalent).

Owner: MPDSC

7.5 University credits

Grading scale: TH - Very good (5), Good (4), Pass (3), Fail

Education level: Advanced level

Main field: Computer Science, Information Technology

Department: 37 - COMPUTER AND INFORMATION TECHNOLOGY

Course opportunity 1

Language of instruction: English

Registration code/opportunity code : 87122

Searchable for exchange students : Yes

Block schedule: D

Maximum number of participants: 160 (at least 10% of the places are reserved for exchange students)

Status, vacancies (updated continuously) : Yes

Points distribution

Module	Lp1	Lp2	Lp3	Lp4	Summer course	Not LP	Exam date
0119 Assignment 7.5 credits Rating scale: TH							

In programs

MPALG Computer Science - Algorithms, Programming Languages and Logic, Year 1 (elective)

MPDSC Data Science and AI, Year 1 (compulsory)

MPHPC High-Performance Computing Systems, Year 1 (elective)

Examiner:

Ashkan Panahi

 [Go to course homepage](#)

Competence

Basic eligibility requirements for advanced level

Applicants with a program registration in a program where the course is included in the program syllabus are exempt from the above requirements.

Special eligibility

English 6

Applicants with a program registration in a program where the course is included in the program syllabus are exempt from the above requirements.

Course-specific prerequisites

A course in programming in a general programming language (e.g. C/C++/Java/Python or similar). A course in mathematics (e.g. analysis, linear algebra, applied mathematical thinking), and a course in mathematical statistics. The course "Introduction to Data Science and AI" (DAT405) or similar. It is strongly recommended that the student has taken a course in Machine Learning, such as DAT340, TDA233 or similar, or that such a course is taken in parallel with this course.

Purpose

The course aims to explain how various well-known AI systems work, provide insight into how to build such systems, and practice developing AI systems yourself. The course takes a broad perspective and also includes related areas such as data science, algorithms, and optimization when appropriate.

Learning outcomes (after completing the course, the student should be able to)

After passing the course, the student should be able to:

Knowledge and understanding

- Provide an overview of various applications within AI and related areas.
- Describe how various well-known AI systems work and are used.
- Explain how AI techniques relate to other types of advanced information processing.

Skills and abilities

- Identify problems that can be solved with AI techniques and other advanced information processing
- Design simpler AI systems for various applications, including model selection and system design.

- Implement AI systems using programming in combination with various tools and programming libraries.

Judgment and approach

- Discuss the advantages and disadvantages of different types of approaches and models within AI
- Reflect on the inherent limitations and opportunities of today's AI methods
- Critically analyze and discuss AI applications with respect to ethics, privacy, and societal impact
- Demonstrate a reflective attitude in all learning

Content

The course teaches AI system design in several different ways:

- Reading articles and lectures that describe various AI systems and their design (e.g. AlphaZero, Watson, self-driving car systems,...)
- Opportunity to test and see the implementation of various simpler AI systems yourself.
- Own problem solving in the form of design and implementation of simpler AI systems.
- Discussion about the possibilities and limitations of AI, ethics and societal impact.

Organization

Lectures and modules with assignments and mini-projects – these are mainly carried out in groups of two people.

Literature

Course literature in the form of articles, etc. which are presented during the course.

Examination including mandatory elements

Assignments and mini-projects. No exam.

The course examiner may examine individual students in a different way than stated above if there are special reasons, for example if a student has a decision from Chalmers regarding targeted pedagogical support due to a disability.